D4. Associations and regression models

Regression estimates how the outcome changes with each predictor, holding the other predictors constant. Dashed boxes are beyond this course: routes for further learning.

Relationship between an outcome and one or more predictors
Predictors (explanatory variables) can be continuous or categorical

What type is the outcome?

Continuous

Correlation (strength only)
Pearson r: linear association
Spearman rho: monotonic, uses ranks

```
cor.test(~ sbp_0 + age, data = hearts)
```

Simple linear regression
Slope = difference in mean outcome per 1-unit difference in x, with 95% CI

```
lm(sbp_0 ~ age, data = hearts)
```

Multiple linear regression
Adjusted slopes; categorical predictors become indicator variables

```
lm(sbp_0 ~ activity_hr + age + sex,
   data = hearts)
```

Binary

Logistic regression
Models the log-odds. exp(coefficient) = odds ratio; adjusted OR (aOR) with several predictors

```
glm(high_bp_0 ~ smoker + age10 + sex,
    family = binomial, data = hearts)
```

Course data: smoking OR 3.39 crude, aOR 2.04 (95% CI 1.17 to 3.54) adjusted for age and sex

Check: enough events per coefficient; linearity of continuous predictors on the log-odds scale. With a common outcome an OR is not a risk ratio.

Count

Poisson regression
Models the log rate. exp(coefficient) = incidence rate ratio (IRR). Different follow-up times enter as an **offset**

```
glm(clinic_visits ~ programme +
  age10 + high_bp_0 +
  offset(log(followup_months)),
  family = poisson, data = hearts)
```

Overdispersion: variance larger than the mean. Pearson chi-square / residual df well above 1 (course data about 2) makes CIs too narrow. Use `quasipoisson` or negative binomial (`MASS::glm.nb`).

Other outcome types

Ordinal (e.g. satisfaction 1 to 5)
Proportional odds (ordinal logistic) regression, e.g. `MASS::polr`

Nominal, more than two categories
Multinomial logistic regression, e.g. `nnet::multinom`

Time-to-event with censoring
Kaplan-Meier curves, log-rank test, Cox regression (hazard ratio), `survival` package

Repeated or clustered observations (any outcome type)
Mixed-effects models or GEE: see D6

Explaining (estimating an association)
Choose adjustment variables from a causal diagram, not from p-values. Adjust for **confounders** (common causes of exposure and outcome), not for **mediators** (on the path from exposure to outcome). Association is not causation.

Predicting
Judge the model on data it has not seen: split into training and test sets (or cross-validate). Avoid **leakage**: no test-set or future information while building the model.

Checks for linear regression
Independence (from the design); linearity; constant spread; approximately normal residuals. Use `plot(fit)`: residuals vs fitted, Q-Q plot. The outcome itself need not be normal.